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Energy enhancement using Multiobjective Ant colony optimization with Double Q learning algorithm for IoT based cognitive radio networks

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ABSTRACT

Internet of Things (IoT) is the efficient wireless communication in the modern era, energy efficiency is the primary issue that focuses mainly on the Cognitive network. Most of the CR networks are focusing on battery powdered to predominantly utilize the data dissipated in terms of spectrum sharing, dynamic spectrum access, routing and spectrum allocation. The clustering and data aggregation are the best efforts technique to enhance the energy modeling. Multiobjective Ant colony optimization (MOACO)and greedy based optimization proposed with Deep Reinforcement Learning with Double Q-learning algorithm. Most of the IoT bed models involve data aggregation and energy constrained devices with optimization techniques to enhance utilization. The cluster-based data utilization is proposed with the Q-learning algorithm and it enhances the inter cluster data aggregation. The network lifetime is improved with AI-based modeling with intra-network to enhance green communication. The simulation experiments showcase that the throughput, lifetime and jamming prediction is analyzed and enhances the energy using the MOACO, when compared to the artificial bee colony and genetic algorithm. The jamming activity at low, high moderate stages is analyzed using the AI and MOACO algorithms.

1. Introduction

Due to the emerging technology and with the advent of the CR network, the environment is easily tackled, CR network involves the formation of a cluster to monitor the node. A cognitive radio can change the best parameters in the transmitter based on the working environment. CR supports a wide variety of technologies to enable the wireless adaptability to dynamic spectrum access with the characteristics of reconfigurability and adaptability [1]. CR is assumed to have the following functionalities to meet out the transmitter needs such as Flexibility and agility it can change the radio parameters to achieve the spectrum that needs sensing and it can measure the environmental state with spectral occupancy. The sensing can be essential for a device to sense the transmitter parameters with the basic knowledge of the Radiofrequency environment. Learning and adaptability can modify the pattern of working nature based on the input behavior with the environmental situation and supports the pre-coded algorithm mechanism with the ease of access to the learning approach. CR can support communications established between two non-CR platforms with different standards without any centralized control. With the emerging growth towards the demand in spectrum usage spectrum policymakers seeks the solutions for spectrum scarcity [2]. Several experimental studies have proven records that most of the licensed users are holding the spectrum in idle usage, hence those spectrums are identified by unlicensed users to make use of that spectrum in that licensed band. This principles approach establishes the dynamic way of accessing the spectrum and flexibility towards it [3]. Multiple access techniques can be deployed in the areas of a higher level of transmission using the same band. Energy efficiency is the major challenging issue as it affects the network lifetime and stability. Scalability is a major factor taken into consideration where the network accommodates a large number of nodes. In CR when the nodes increase the performance of the network and broadcast communication should also be monitored as shown in Fig. 1. This may lead to a lack of energy for all the nodes to sustain the operational parameters. Scalability with energy preservation is a major challenging issue in CR [4].

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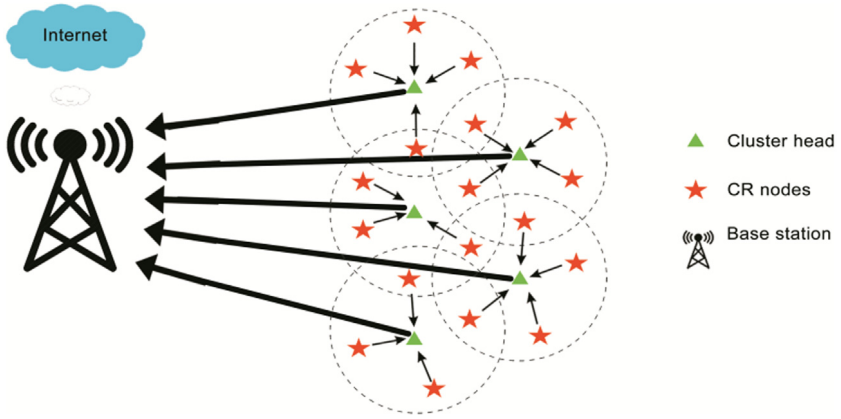


Fig. 1. Cluster based CR network.

CR network is a futuristic technology to solve the demand for spectrum shortage and proper utilization of the spectrum from primary users (PU). While Cognitive users also termed as secondary users (SU) may approach the PU for the utilization of idle spectrum for efficient transmission and enhanced energy-based data communication [5]. Co-operative spectrum sensing analyses the information from multiple SU to identify the spectrum usage and utilization from PU. The received signal strength in PU might be disrupted by interference by multipath fading. This fading can be overcome by different SU placed in different geographical locations and each individual SU will obtain separate sensing information that has been combined to reduce uncertainity [6]. This cooperative sensing technique obtains an optimized better decision results to attain the spectrum occupant information and it may enhance the energy to be maintained in all nodes in equal. This optimal decision of a cooperative scheme reduces the misdetection and false alarm probability. The cooperative spectrum sensing enhances the energy of nodes by making the communication in three paradigms [7]. Centralized Spectrum Sensing, **I**n this sensing each SU will update their sensed information to a secondary base station. The station will analyze the spectrum holes and allocate the channel based on the fusion rules associated with the sensing information. When more SU's are wavering in a particular geographical region the information will be gathered infusion center and analyze the information based on it and overhead might increase in dealing it [8]. Distributed Spectrum Sensing, Each SU will share their sensed information with each other about the spectrum transmission in this sensing. This sensing does not need an infrastructure to support the sensing information. This sensing cannot achieve the capacity of the centralized manner in information sharing [9].

In External Sensing **t**he sensing can be performed with the help of an external agent in the SU. The centralized and distributed sensing are assumed to be of internal sensing due to its operational parameters. The cooperative sensing rather than concentrating on multipath fading and shadowing, it enhances the energy of the nodes within the desired level [10].

CR network provides a random access environment in which the users operate on a channel that is not being used by the primary users in a licensed manner [11]. The security paves the characteristics of the CR network having the attack such as jamming, eavesdropping, selfish behavior, an intruder. The proposed work provokes an energy reservation during the jamming attack mitigation with the support of the MOACO algorithm as shown in Fig. 2. The jammer has been induced in the common control channel and the primary user data can be hijacked with this activity. The secondary users share the data with the neighboring secondary users regarding the report collected from the central fusion center. The secondary users share the data which has been highjacked without knowing the jammer activity and the central fusion report is a centralized work. The MOACO algorithm uses the technique to identify the data in transmitter and data path routing without the attack getting acquainted with it from the fusion center [12]. The channel condition is estimated with the received signal strength is measured by primary user availability and the path of routing is based on IoT source destination set in CR network manners to provide an optimal solution [13].

Clustering is a common approach to deal with a huge number of nodes in a network. In CR networks cognitive users are allowed to form a group based on the geographical assumption or operational parameters [14]. The Cognitive user forms a group and elects a head within the group termed as cluster head. The cluster head will collect the sensing information from all the secondary users within the cluster. The cluster head report those information to the Fusion center on behalf of their members. The cluster head which are farther away from the fusion center will share the information to the nearest cluster head and all those sensing information will be communicated to the fusion center in a periodical time interval, it may reduce the energy consumption in a frequent time slot during the reporting stage of the CR networks. After every cooperative sensing process a global optimal decision will be reported to the Fusion center and it enhances the energy consumption within the CR networks. In CR network, the locally collected information at FC and broadcasting the information will have more impact on security issues and it is a bit challenging to overcome the issues [15]. The potential risk a CR network faces are the vulnerable challenges for the network to overcome and makes the communication error-free. The risk involves Receiver jamming, In this attack, the attacker reduces the threshold level of the SNR in the particular channel. The threshold level when reaches a lower value the network invokes a anonymous changes and it may affect the network performance. This impact makes the attackers to hijack the information in PU and occupies the spectrum. CR network are prone to attack and have more risk in comparison with the other wireless network, the paper focus the jamming attack that interrupt the CR network data transmission. The proposed model outline the attack mitigation with the energy enhancing in each hop of the channel with the IoT source to destination pair accordingly. A double Q-learning mechanism has been used to learn channel conditions in a dynamic channel selection scheme and it is a unsupervised and online artificial intelligence technique which helps to improve system performance through simple modeling and learning. The RL model has been proposed to improve the channel consumption in a dynamic environment for all SUs [16].

Ant Colony Optimization (ACO) [17,18] is a metaheuristic algorithm where the optimization problem are being depicted in graph in which the ant may obtain optimal solution. Whenever ant obtain solution in the trailing path with the help of the search space the ant obtain complete solution with in the surrounding [19,20]. The path

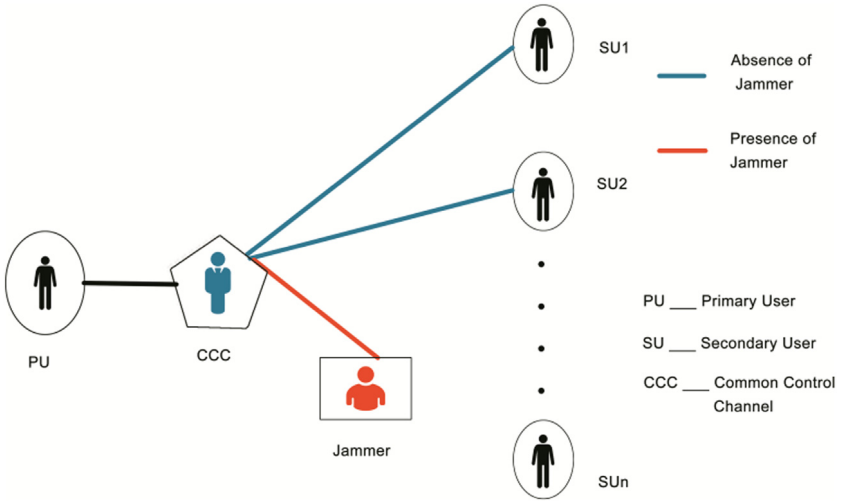


Fig. 2. Jammer induced CR network.

routing in CR network maximize the node energy and if even the attack has intruded the network the cluster node will get drained with the energy consumption and the optimal solution has been obtained with the multi-objective nature [21]. The MOACO has been used to obtain qualitative information with the support of heuristic information hence forth the algorithm has been utilized well in the energy analysis path and the energy conservation in CR node toward data traveling [22,23]. In MOACO algorithm each iteration has been performed with the ordered way in CR connected node [24]. In each iteration the ant pheromone is probabilistically applied to the node wise distribution to conserve more energy even the node information is hijacked. The transition rule is framed with the Q-learning algorithm to identify the ant exploration in the objective value with the amount of pheromone deposited in each path. The MOACO algorithms has been identified and path estimation with energy monitoring is performed with the objective of enhancing the solution in characteristic of heuristic nature and pareto archive to establish the solution in the CR network when compared to ABC and genetic algorithm

Contribution: The main contributions of this study are

(i) First of all, the energy monitoring during the jamming attack mitigation is predicted in CR-IoT edge network using MOACO.
(ii) The comparative analysis of the genetic algorithm and other metaheuristic algorithm is performed with the inter cluster data model using Qlearning algorithm to focus the increase in the network lifetime.
(iii) An hybrid model has been proposed for enhancing the energy monitored in CR node using the greedy algorithm for making the data gathered using the cluster node.
(iv) A wide-ranging analysis has been done to enhance the energy monitoring and to evaluate the proposed model.

The rest of the paper is organized as follows. In Section 2 Literature analysis has been performed. Section 3 deals with the jamming attack in the CR network model and the energy conservation using MOACO algorithm during attack is discussed with the mathematical formulation, Section 4 deals with the path selection using MOACO. Experimental setup and result are described in Section 5. The concluding remark and future research is dealt in Section 6.

2. Literature survey

Dorigo et al. proposed that the problem-solving techniques have been given to form a domain orientation with the multiple objective optimizations in the stochastic and the population based algorithm to solve the domain orientation in the ACO community [2]. The multi-objective communication has been done with the ACO in terms of the success or failure approach in the assignment of the success in terms of the number of possible ways to determine the ACO algorithm. The multidimensional ACO has been determined by the focus of extension of the single objective.

Stützle et al. gave an outline that the ACO algorithm in the ant colony system depends on the MAX–MIN Ant system with MOACO algorithms. The MOACO algorithm has been summarized with the number of pheromone matrices and heuristic matrices. The proposed methods enhance the analysis of the MOACO algorithms in the similar problems domain. The MOACO algorithms have the objective of fulfilling the multiple objective traveling salesman problems. The paper focuses on the objective traveling salesman problem with the analysis of problem domains, the analysis interprets the modification of MOACO with the pheromone model in the solution construction for the Pareto optimal solutions in the metaheuristic applications [25].

Adeliet al. proposed that the Secondary users can harvest the energy within the node signal or from PU's based on the received signal strength. If the signal strength is good in nature it harvests the energy in the SU itself and battery of each SU's can be charged [26]. The cooperative sensing may be applied in SU which optimizes the energy harvesting based on time constraints. Based on the energy harvesting performance the SU's transmission rate and sensing time can be enhanced.

Althunibat et al. proposed the SU need more energy consumption due to the continuous searching of primary user without causing interference, as more number of SU's are involved in this process, it is predominant to focus on collaborative spectrum sensing with the PU and SU's, which may automatically increase the energy consumption with this sensing. The energy conservation on collaborative sensing makes the minimum false alarm probability and misdetection ratio [27]. The clustering approach in the SU's with the collaborative sensing makes the energy conservation to the next paradigm.

Song et al. highlighted that the Secondary user senses the channel based on the sensing information collected collaboratively with PU. Each SU submits the sensed information in regular time intervals to the Fusion center to obtain a sensing decision. The Fusion center applies a majority rule with the collected information about the total number of PU available in that particular time slot. SU submits the sensed information as '0' if PU is not present and '1' if PU is present. This information collected at the fusion center highlights the channel and spectrum usage relevant to energy monitoring of the particular SU. If

a malicious SU intrudes in the network it may drain the entire energy of the particular node. This issue is a challenging task of maintaining the energy level within the CR network [28].

Yang et al. proposed that the clustering is a common approach to deal with a huge number of nodes in a network. In CR networks cognitive users are allowed to form a group based on the geographical assumption or operational parameters. The Cognitive user forms a group and elects a head within the group termed as cluster head. The cluster head will collect the sensing information from all the secondary users within the cluster [29]. The cluster head reports that information to the Fusion center on behalf of their members. Kozal A. et al. outlined that the cluster head which is farther away from the fusion center will share the information to the nearest cluster head and all those sensing information will be communicated to the fusion center in a periodical time interval, it may reduce the energy consumption in a frequent time slot during the reporting stage of the CR networks [30]. Wei & Zhang proposed that after every cooperative sensing process a global optimal decision will be reported to the Fusion center and it enhances the energy consumption within the CR networks [31].

The IoT based systems are complex system which follows the distributed system. Swarm intelligence are self organized and distributed with the dynamic approach that has been utilized with the multiple IoT system [32]. The energy consumption of the node has the biggest routing in wireless sensor networks due to the battery limitation and the ant colony optimization has been used for routing the data with life time aware routing algorithm with the objective function to evaluate the new pheromone in the routing distance. The result of the multiple simulation has the ability to enhance the balanced transmission in the system among the node and the energy consumption is used to extend the network lifetime [33]. Fei Li et al. proposed that the complex environment utilize the highly equipped WSN to enhance the long time service and the highly qualities data transmitted to the center. The traditional routing algorithm enhance the WSN with the Basic Ant-Based Routing (BABR) to identify the shortest path and the convergence in a slowly occupied manner that outperform the local optimum in the algorithm. Quantum Ant Colony Multi-Objective Routing has been proposed in the manufacturing environment to identify the quantum computation within the routing algorithm in which the quantum bit re used for the node pheromone in the search path for the routing to be performed within the load balancing degree of the node to focus the energy consumption, network life time and load balanced structure [7].

Srinidhi et al. proposed that the data transfers are happened between the edge device with the resource contained in the network constitute a IoT framework [34]. Large amount of data are being overloaded within the network to monitor the network with the issue in the network routing, congestion, energy conservation, reliability, quality of service to provide network optimization [9]. Proposed that the spectrum sensing method in cognitive radio based on the fuzzy membership functions which are quantized using a near-optimal quantization scheme with low computational complexity. A membership function is obtained in a Secondary User (SU) using an ordinary energy detection method and its output is sent to a Fusion Center (FC) after quantization through a noiseless reporting channel, where the final decision is made. The FC combines the received information using some fuzzy rules such as algebraic product to make a final decision about the absence/presence of a primary user [35].

Anal Paul presents a cluster-based approach at the fusion center since it handles a large amount of data and fuzzy based c-means clustering and it has been decided at energy based cooperative spectrum sensing has been analyzed. The projection of linear based problems at the data set patterns has been proposed. The performance of Fuzzy c means clustering degrades at low signal-to-noise ratios (SNR) and in the presence of multiple PUs as energy data patterns at the FC are often found to be overlapping. To address the problem, the scope of kernel fuzzy c-means (KFCM) on energy detection based CSS through the projection of non-linear input data to a high dimensional feature space. Meiqin Tang Proposed that the trade-off between utility and energy consumption in orthogonal frequency division multiplexing based CR network is investigated [36]. Energy efficiency problem is very important in the field of CR network, where the utility is maximized and the energy consumption is minimized in such a CR network. An improved particle swarm optimization (PSO) algorithm makes the optimization problem PSO employs a co evolutionary methodology, and then divide-and-conquer strategy provides energy efficient feasible solution [37].

Faroq Ali proposed an approach for joint optimal design system parameters is considered for energy efficient cluster based spectrum in CR networks [38]. An optimal iterative algorithm is proposed to determine the sensing time, data transmission time and number of CR maximize the energy. An iterative algorithm with low computational complexity is proposed to determine joint optimal sensing time, data transmission time and the number of CR users [39].

Chen K-C proposed a jamming attack where malicious SU makes the jamming in the PU. It causes the interference in CR network, malicious SU moves from one geographical region to another within the network. Also, Proposed a false feedback attack where malicious SU are grouped to form a group and it makes another group of SU to violate the network policy. This type can happen at both centralized and distributed networks. Proposed a common control channel where the attacker generates the periodical pulses in the network [40]. The jamming happening in one block of channel will affect the other channel too residing in the network. Leon et al. presented that the Radio parameters has lot of control and it enhances the objective with center frequency, bandwidth, power, modulation, bandwidth [41]. The optimal solution in the method is to secure communication, low power and high rate. Proposed a spectrum sensing data falsification attack sends the false information in local sensing to the central collector. The fusion center makes the wrong decision on the collected information within the network [42].

Jin et al. Proposed a congestion attack is a nature of attack and attackers flooded common control channel with a extended DDOS attack [43]. This type of attack may cause the DoS attack at the physical layer. There is no modification of the incumbent signal with the primary transmitter being located at the malicious parameters. The performance has been monitored with the three phases such as verification of signal characteristics, signal characteristics at the receiver, localization of the channel within the network. Minho Jo et al. presented that a selfish CR occupies a part of the resource of multiple channels which may deny the access of other nodes in the CR network. COOPON an attack detection scheme with multichannel resources by cooperative neighbor cognitive radio has been identified [44]. This selfish attack is a model of denial of service attack with cross layer service protocol called ADSP has been used for dynamic CR networks [45]. This attack makes the CR network to face the lower delay, higher throughput and better delivery ratios to be attacked.

The IoT provide connectivity that is ensured with the energy harvesting and the cognitive radio technique, the energy depletion may be due to worst channel condition and the battery power. Krishna et al. proposed that the Energy enhancement is a major criteria has to be fulfilled in the CR network with IoT, the IoT [46–49] has been proceed with the five layer in CR network such as sensing, communication, storage, processing and application building [50]. The significance of cross layer energy optimization is presented with the different implementation, model based review has been carried out in the IoT environment [51]. Yuan Hong Li proposed that the IoT has the resource distribution is done in space that decide the precision ratio with poor clustering indexing in IoT with the improved data fusion in the optimization for the IoT in the clustering in the improved ant colony algorithm in the proposed manner. The algorithm for the cluster for resource indexing Is mapped with the directed graph and hence the

semantic feature extraction in the cluster resource, storage information with the IoT [52]. The ant colony algorithm captures cluster information with the velocity, position and the clustering function to target the optimization parameter with the indexing option [53].

3. Energy conservation model using a cluster approach and jamming impact in network model

Consider a network model with a multihop IoT network with each node having CR capability. This setup can access the free channels which are not assigned and the CR IoT network has a set of PR channels as 'c' with the bandwidth 'B'. Consider a channel A in C for a CR IoT transmission. ON/OFF has an ON state busy and busy transmission is marked as $T_{ON.}$ The probability distribution of the transmission has been established with the function for forming the $F_{TON}[t]$ and the function of forming the off-state condition in the channel is determined by the parameter as $F_{TOFF}[t]$. The CR network is assumed to be off the IoT device connectivity in CR networks has the transceiver that cooperates with the neighboring CR IoT devices. For each hop 'h' and the given path 'p' has a set of available channels $c_{P,}$ H ∈ C, and the threshold is assumed to be lower in nature and the data path length is assumed to be of 'L'. The remaining channel is assumed to be of $c_{P,H}$ and hence the data jamming behavior is assumed to be off in the channel quality as shown as $S\left(T_{jam^i}\right)$ is assumed to be on the jamming timing in the network and hence the data path travel is assumed to be on the condition in the c Rayleigh path fading channel where S denotes the Shannon coding channel allocation for the path [54]. The path has been accepted as the with the Rayleigh path fading channel having the routing in terms with $P(R_{ph})$.

Consider a multidimensional field with N CR node and the free space environment is denoted by M^2. In a multipath fading channel has the packet transmission with multiple hops with the power loss M^4.

$E_{Txn}[D,\ S]$ where D represents the multiple hop distance and the S denotes the size of the data packet.

$$E_{Txn}[D,S] = \begin{cases} S_{E_{Txn}} + S_{D_{FsE}} M^2, M < M_0 \\ E_{Txn} + S_{P_{EL}} M^4, M \geq M_0 \end{cases}$$

D_{FsE} represent the free space distance, P_{EL} is the multipath loss, M_0 represent the threshold

$$M_0 = \sqrt{\frac{D_{FsE}}{P_{EL}}} \tag{1}$$

3.1. Cluster head formation

The cluster head formation has been done with the most life time predicted secondary users and the cluster head selection is based on the threshold selection with the corresponding nodes.

$$\text{Threshold(i)} = \frac{\rho_{opt}}{1 - \rho_{opt}(r \cdot mod\left(\frac{1}{\rho_{opt}}\right))} * \frac{C_{E(r)}}{C_{avgE(r)}} \tag{2}$$

$C_{E(r)}$ represent the current energy in a cluster

$C_{avgE(r)}$ represents the average energy in a cluster

$C_{avgE(r)} = \sum \frac{C_{E(r)}}{N}$ for every node i

ρ_{opt} represents the optimization with respect to the jamming attack in the cluster node

$r \cdot mod$ performs the modulation of cluster head node with the optimized node with r *times*

3.2. Multiobjective ACO

(a). Initialize the cluster head

(b). Let CH = Ant head

(c). Let $D = CH_i$ // measure the distance traveled by each ant i

(d). $N_{CH} = D_i$ // The ant framework estimated with the $N_{CH.}$

(e). $D_i = p+P$ // The pheromone deposited is calculated t the distance is predicted 'D' with the sum of the journey with 'p' price with p value.

(f). $CH_i = P_{h,k}+D$ // The following ant in the cluster has the deposit feedback and the path has been followed in response with the predicted ant system

(g). The ant movement is predicted with the distance covered with the distance crossed by each ant in different cluster heads that is from cluster to $cluster_j$. Where $i \neq j$. The pheromone i assumed to be of the simple ant and the measure is calculated using the below equation as shown in Eq. (3)

$$P_{h,k=\frac{(A_{h,k})^{\alpha}+(H_{r,f})^{\beta}}{\Sigma(A_{h,k})^{\alpha}(H_{r,f})^{\beta}}} \tag{3}$$

where $A_{h,k}$ represents the amount of pheromone deposited from the $cluster_i$ to $cluster_j$. $H_{r,f}$ represents the objective function which is inversely proportional to the energy between the $cluster_i$ to $cluster_j$. α represent the pheromone estimated in the path and the β represent the heuristic nature of the ant.

(h). The path has been estimated with the condition that

$$P_{h,k=\frac{(A_{h,k})^{\alpha}+(H_{r,f})^{\beta}.CHi.Di}{\Sigma(A_{h,k})^{\alpha}(H_{r,f})^{\beta}}} \tag{4}$$

measure the $P_{h,k}$ that the pheromone is updated in the path

else

$P_{h,k} = 0$

End

(i). Estimate the euclidean distance between $cluster_i$ to $cluster_j$

$$P_{ED}=\sqrt{(CL_i.ax - CL_{j.}ay)^2 + (CL_i.bx - CL_{j.}by)^2} \tag{5}$$

Here, ax and by represent the coordinate of the cluster formation and P_{ED} denotes the probable euclidean measurement

CL_i *represents the cluster formation group* '*i*'

CL_j *represents the cluster formation group* '*j*'

(j). $P_{h,k}$ is updated whenever the ant reaches the pheromone deposition.

(k). The pheromone is updated every time and the measurement is estimated with the edge of the cluster.

$$P_{h,k} \leftarrow (1 - P)\, A_{h,k} \tag{6}$$

(l). The cluster head will not choose the artificial ant since it may eradicate the ant and the pheromone gradually decrease.

$$(m).\ A_{h,k}\left(t_n\right) = PA_{h,k}(t) + \Delta A_{h,k} \tag{7}$$

(n). $A_{h,k} \leftarrow A_{h,k} + \sum_{s=1}^{n} \Delta A_{h,k}{}^{s}, \forall(h,k) \in L$, Where the ant passes the cluster and reaches the pheromone deposition. $\sum_{s=1}^{n} \Delta A_{h,k}{}^{s}$ represent the edges visited during the travel of the ant.

(o). $\Delta A_{h,k}{}^{s} = \begin{cases} 1/LE \\ 0 \end{cases}$ where LE represents the length of the edge.

(p). The minimum path has been estimated with the ant solution and the pheromone is utilized.

3.3. Deep reinforcement learning with double Q-learning model

The Q-learning model estimates the approach with the off-policy control algorithm where the policy agent is used to update the following condition for making the cluster head to be applied to the behavior environment [55]. The time interval for the measured environment is observed by the knowledge of each agent and the Q learning for the knowledge measured is denoted by the

$$Q^{a}_{n+1}\left(s^{a}_{n}, a^{a}_{n}\right) = (1-€)\,Q^{a}_{n}\left(s^{a}_{n}, T^{a}_{n}\right) + €[r^{a}_{n+1}\left(s^{a}_{n+1}\right) + ¥ \max_{a€A} Q^{a}_{n}(s^{a}_{n+1}, X)] \quad (8)$$

where s^{a}_{n} represents state of agent a at time t, a^{a}_{n} represents action, $r^{a}_{n+1}\left(s^{a}_{n+1}\right)$ represents delayed rewards for the action taken at time n and received at time n+1. The learning rate I predicted with the $€(0 < € < 1)$ is the learning rate and $(0 < ¥ < 1)$ represents the cut off factor.

Initialize the Primary CR network with Q_1 and target CR network with Q_n
Time limit t≪1 // The Buffer in the CR network i assumed to be of B with the time limit
Do Iteration //For each iteration do
  //For each operating environment do
select $g_n \sim \pi(g_{n,} s_n)$// Examine the state s^{a}_{n} and Execute g_n and predict the next state $s^{a}_{n}+1$
Reward $r_n = R_W(s^{a}_{n}, g_n)$
Store $(s^{a}_{n}, g_{n,} R_{W,} s^{a}_{n+1})$ primary network
Store in buffer B
update and do
Sample $SP_n = (s^{a}_{n}, g_{n,} R_{W,} s^{a}_{n+1}) \sim B$
Predict the target in CR network with Q_nValue
$(Q_1(s^{a}_{n}, g_n) - Q_n(s^{a}_{n}, g_n))$ // Perform gradient descent step on
$Q_n \leftarrow t^{*}\ Q_1(1-t)\ ^{*}\ Q_n$// Update target network parameter

4. Greedy mechanism-based PSO for path selection

The greedy algorithm predicts the choice for the optimal solution in a reasonable time. The optimality prediction depends on the previous decision, not on the current aspects or future aspects that adhere to the subproblems. The greedy process is a subset of the dynamic programming aspects [56]. The decisions are considered to be of the optimality in the precise approximation for the optimum identification [57]. The Traveling salesman problem is identified based on the efficiency of the timing constraint parameter in the immune algorithm for the global parameter to identify the delete cross operator and path over to the network in the greedy allocation [58]. The CR network long with the IoT edge device support the primary and secondary user in the network for greedy based network allocation and hence the CR network is supported by the IoT CR network as shown in Fig. 3.

Particle swarm optimization enables the solution for the particle to identify the stored record which obtains the optimal solution for the best particle [59]. The particles are identified with the objective function framed with the fitness value as a pop particle [25]. The fitness value is estimated with the $F_{Particle}$ and the corresponding population is assumed to be of the $P_{Particle.}$ The objective function is assumed to be of the minimum path cost and the velocity is done by the random generation in the target value, $F_{Particle}$ and the termination in the target condition in the PSO [60].

(a) The data represented with the global solution is the $F_{Particle}$
(b) The Velocity is the total average data
(c) $P_{Particle}$ is the optimal solution for the particle associated with the CH.

1. The particle is assumed to be of the cluster head and it mentions the dimension with position and the average length traveled.

2. Random distribution is applied based on the size of the population.

3. The path distance is measured based on the fitness value in the nodes with the Euclidean distance.

$$PD = \sqrt{(x_i - x_j)^2} + (y_i - y_j)^2 \quad (9)$$

where $\sqrt{(x_i - x_j)}$ represent the position value for the corresponding nodes. The random value generates the minimum value for the fitness in the best particle [61].

4. The new particle is formed from the previous set fitness value to reach desired outcome in the particle estimation with the minimum aggregation.

(i) The velocity is obtained in the particle position with the respect to the rate and the particle position is changed accordingly in the given manipulation as shown in Eq. (10)

$$Vel_N = \rho * Vel_{old} + \rho_{1(FParticle - CP\ Particle)} + \rho_{2(FParticle - CP\ Particle)} \quad (10)$$

where the ρ represents the weight calculation in term of the PSO with the positional value.

(ii) The new position is estimated with the old value and the velocity in the PSO node as shown in Eq. (11)

$$Position_N = OLD_N + Vel_N \quad (11)$$

Hence the new particle arrives with the above measured velocity and position.

5. The fitness value has been estimated with the new value of the position

6. The next path identification is done with the comparison of the new and old fitness value and hence the new position identifies the measured iteration process.

if $FN_{NEW} < FN_{OLD}$
  $FN_{OLD} = FN_{NEW;}$
else
The previous article is assumed to enter the next process
End
where FN_{NEW} represents the new fitness value.

7. The iteration in the maximum fitness value obtained is assumed to be of the best fitness and it will be the best inter cluster aggregation path.

5. Experimental setup and results

The investigation has been done with the primary user activity under different levels of jamming performance. The levels of jamming attacks have been done in comparison with the proposed MOACO algorithm and the activity varies from low to high (0.11–0.89). The packet delivery ratio also increased in the proposed method that has been compared with the general genetic algorithms and artificial bee colony algorithm with high probability and moderate different jamming levels it shows that as the availability of the channel decreases and the network performance increase when the P_B increase with the gain as shown in Figs. 4–6. The simulation setup has been shown in Table 1.

The jamming attack has been measured with the various level of Primary user and colluding user in the impact with the low P_B is estimated with the 0.1 and the high is estimated with the 0.9 and the moderate is estimated with the 0.5.

Fig. 4 shows that the high attack may make the network to be degraded and hence the data is assumed to be of the and the proposed Q Learning technique mitigate the attack in proportionate way. When

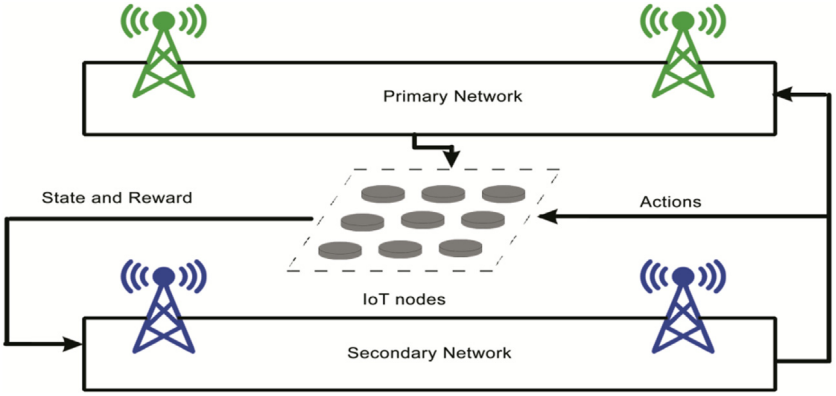


Fig. 3. Jammer induced CR network.

Table 1
Simulation setup.

Parameter	Value
Simulation area	250 × 250 m
Frequency band	700 MHz
Nodes	150
Probability	0.1
Channel bandwidth	4 MHz
Duration	5, 90, 20, 5, 35, 40, 90, 5, 35, 30 ms
Jamming duration	5, 0.4, 5, 2.1, 9, 5, 1, 2.8, 15, 0.1 ms
Cognitive RadioIoT threshold	4 dB
Packet size (Control)	150 bit
Packet size(Data)	4 Kbytes
Energy during transmission	0.1
Transmitter energy	40×10^{-8}
Receiver energy	40×10^{-8}
Maximum lifetime	6×10^{-8}

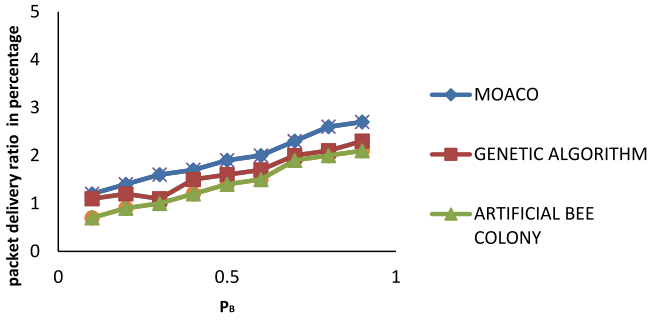


Fig. 4. High Jamming activity.

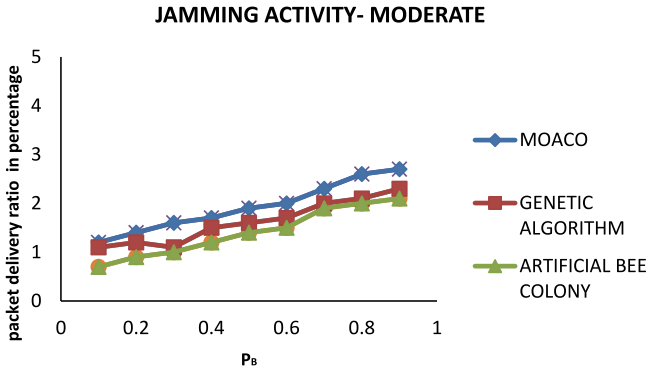


Fig. 5. Moderate Jamming activity.

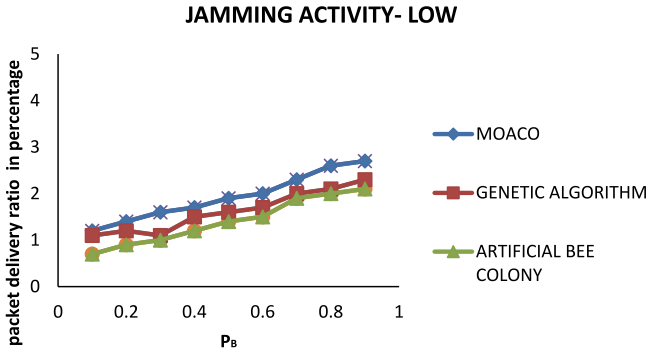


Fig. 6. Low Jamming activity.

the attack level is degraded the network performance is assumed to the range in the lower impact and the jamming is assumed to be of higher between the common channel and the packet delivery ratio increases with the low primary activity and hence it has been decreased.

Figs. 5 and 6 show that impact of the transmission power level in different primary user activities and the different jamming attack has been predicted in the different level. The power factor for the data transmission varies from 0.4 to 1.1 W, when the transmission level increases it automatically enhance the power level. When the power increases the transmission, the time of transmission decreases with the required level and hence the channel availability due to the jamming attack become less.

The performance has also been evaluated with the parameters such as constancy period, lifetime of the network, harvested energy and average energy. The network parameter has also been established with the various CR nodes to 150 nodes. The throughput is measured with the number of packets which has been successfully transmitted in the sink module shown in Fig. 7. The proposed method significantly improves the threshold adaptation. The residual energy has been performed with the last node observed in the CR network area shown in Figs. 8 and 9. The maximization is observed with the comparison with the metaheuristic algorithms such as genetic and artificial bee colony algorithm. The simulation has been performed with the metric bed on stability period, average energy with the throughput with 150 CR nodes. The CR nodes are arbitrarily deployed within the node parameter coverage area of 250 × 250 area with base station for the analysis. Throughput is measured for the packets transferred to the sink from the source. Fig. 7 indicated the throughput performance and it has clearly indicated that the proposed technique has been significantly improved. The throughput using MOACO has been compared with the GA and ABC, the result implies that the MOACO algorithm given significant performance as shown in Table 2.

Table 2
Throughput analysis with MOACO.

Nodes	Proposed MOACO (J)	Genetic algorithm (J)	Artificial bee colony (J)
50	300	150	180
100	378.99	250	300
150	556.98	300	450
200	625.36	450.55	560.39
250	698.98	570.55	620.98
300	725.36	600.25	650.39
350	753.69	625.33	675.9

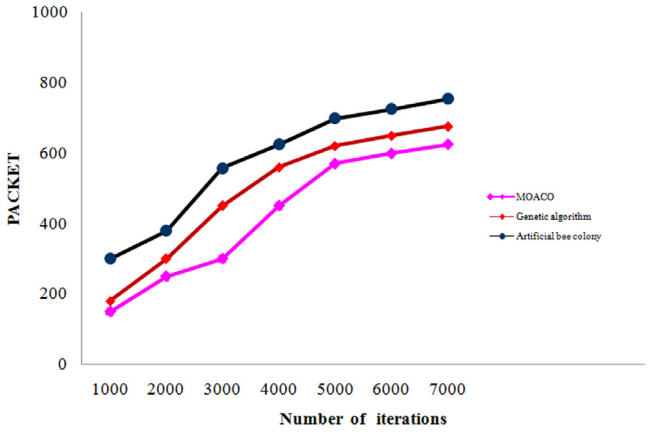


Fig. 7. Throughput analysis in the proposed model.

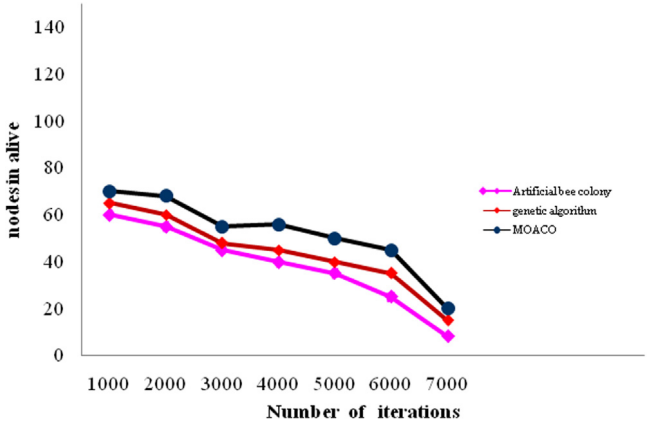


Fig. 8. Network lifetime in the proposed model.

Fig. 8 indicated the Network life time performance is compared with the node in live condition even after the attack has been happened and it has clearly indicated that the proposed technique has been significantly improved. The throughput using MOACO has been compared with the GA and ABC the result implies that the MOACO algorithm given significant performance.

The average energy is measured with the time taken by each node in the network to obtain the maximum duration a CR node live in the network. Fig. 9 indicated the energy performance and it has clearly indicated that the proposed technique has been significantly improved. The throughput using MOACO has been compared with the GA and ABC the result implies that the MOACO algorithm given significant performance. The energy is consistent in the proposed model is gradually increased with the proposed model as shown in Table 3. The through put analysis in the proposed model has been observed that it has been increased by approximately 4.9% compared with the other algorithm.

Table 3
Energy comparison with MOACO.

Nodes	Proposed MOACO (J)	Genetic algorithm (J)	Artificial bee colony (J)
50	70.6	60.6	65.2
100	68.2	55.8	60.4
150	55.3	45.9	48.6
200	56.4	40.4	45.5
250	50.2	35.3	40.4
300	45.4	25.3	35.3
350	20.6	8.2	15.4

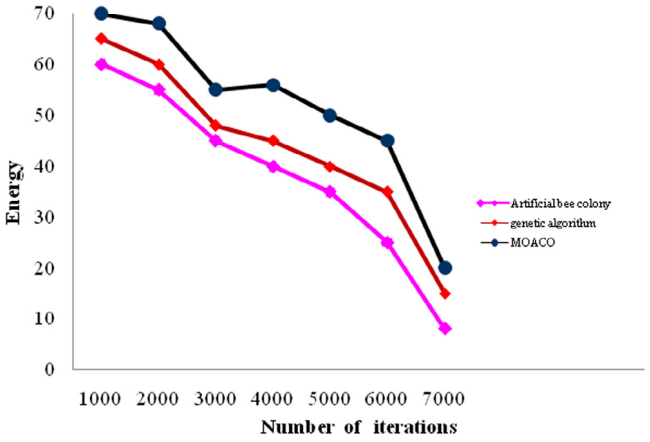


Fig. 9. Residual energy in the proposed model.

Besides handling the residual energy there may be some chances of the drawbacks arising with the MOACO algorithm, which may be overcome with the high computational servers to manipulate the data.

6. Conclusion

The work proposes a hybrid model of mitigating the jamming and energy monitoring using the MOACO model along with the double Q-learning model. The model has been proposed with the IoT model that involves data aggregation and energy constrained devices with optimization techniques to enhance utilization and the network throughput with energy enhancement. The MOACO algorithm uses the technique to identify the data in transmitter and data path routing without the attack getting acquainted with it. The channel condition is estimated with the received signal strength is measured by primary user availability and the path of routing is based on IoT source destination set in CR network manners to provide an optimal solution. The experimental analysis shows the comparison with the genetic algorithm and another meta-heuristic algorithm that enhances the network lifetime, residual energy and lifetime parameters. The proposed model predominantly works on the model to enhance the energy and it is attained. The proposed model can be analyzed with the energy parameters in terms of using traditional and advanced machine learning techniques which may takes more complexity towards execution in the jamming attack, studies may be enhanced to deal with the energy and security parameters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

S. Vimal: Writing - original draft, Writing - review & editing, Conceptualization, Data curation, Validation. **Manju Khari:** Conceptualization, Formal analysis, Supervision. **Rubén González Crespo:** Conceptualization, Formal analysis, Writing - original draft, Writing - review & editing, Supervision. **L. Kalaivani:** Conceptualization, Formal analysis, Supervision. **Nilanjan Dey:** Formal analysis, Writing - review & editing. **M. Kaliappan:** Conceptualization, Data curation, Validation, Supervision.

References

[1] S. Vimal, L. Kalaivani, Kaliappan, Collaborative approach on mitigating spectrum sensing data hijack attack and dynamic spectrum allocation based on CASG modeling in wireless cognitive radio networks, M. Cluster Comput. (2017) http://dx.doi.org/10.1007/s10586-017-1092-0.

[2] Marco Dorigo, et al., Ant colony optimization, IEEE Comput. Intell. Mag. 1 (4) (2006) 28–39.

[3] s. Vimal, et al., Secure data packet transmission in MANET using enhanced identity-based cryptography, Int. J. New Technol. Sci. Eng. 3 (12) (2016) 35–42.

[4] A. Rubio-Largo, M.A. Vega-Rodrguez, Applying MOEAs to solve the static routing and wavelength assignment problem in optical WDM networks, Eng. Appl. Artif. Intell. 26 (5–6) (2013) 1602–1619.

[5] E. Mariappan, M. Kaliappan, s. Vimal, Energy efficient routing protocol using Grover's searching algorithm using MANET, Asian J. Inf. Technol. 15 (24) (2016).

[6] Q. Gao, Y. Zuo, J. Zhang, X.H. Peng, Improving energy efficiency in a wireless sensor network by combining cooperative MIMO with data aggregation, IEEE Trans. Veh. Technol. 59 (8) (2010) 3956–3965.

[7] Fei Li, Min Liu, Gaowei Xu, A quantum ant colony multi-objective routing algorithm in WSN and its application in a manufacturing environment, Sensors(Basel) 15 (2019) 3334.

[8] S.S. Ilango, S. Vimal, M. Kaliappan, et al., Optimization using artificial bee colony based clustering approach for big data, Cluster Comput. (2018) http://dx.doi.org/10.1007/s10586-017-1571-3.

[9] N.N. Srinidhi, S.M. Dilip Kumar, K.R. Venugopal, Network optimizations in the internet of things: A review, Eng. Sci. Technol. Int. J. 22 (1) (2019) 1–21.

[10] N. Kannan, S. Sivasubramanian, M. Kaliappan, s. Vimal, A. Suresh, Predictive big data analytic on demonetization data using support vector machine, Cluster Comput. (2018) http://dx.doi.org/10.1007/s10586-018-2384-8.

[11] R. Geetha, S. Sivasubramanian, M. Kaliappan, et al., Cervical cancer identification with synthetic minority oversampling technique and PCA analysis using random forest classifier, J. Med. Syst. 43 (2019) 286, http://dx.doi.org/10.1007/s10916-019-1402-6.

[12] M. Dua, R. Gupta, M. Khari, et al., Biometric iris recognition using radial basis function neural network, Soft Comput. (2019) 11801–11815, http://dx.doi.org/10.1007/s00500-018-03731-4.

[13] Suresh Annamalai, R. Udendhran, S. Vimal, An intelligent grid network based on cloud computing infrastructures, in: Novel Practices and Trends in Grid and Cloud Computing, 2019, pp. 59–73, http://dx.doi.org/10.4018/978-1-5225-9023-1.ch005.

[14] M. Khari, A. Sinha, E. Verdú, et al., Performance analysis of six meta-heuristic algorithms over automated test suite generation for path coverage-based optimization, Soft Comput. (2019) http://dx.doi.org/10.1007/s00500-019-04444-y.

[15] Y. Lu, W. Hu, Study on the application of ant colony algorithm in the route of internet of things, Int. J. Smart Home 7 (3) (2013) 365–374.

[16] W. Jinyi, Y. Qin, A. Shrestha, S.-J. Yoo, Optimization of cognitive radio secondary base station positioning and operating channel selection for IoT sensor networks, in: International Conference on Information and Communication Technology Convergence: ICT Convergence Technologies Leading the Fourth Industrial Revolution, ICTC 2017, 2017, pp. 397–399, doi:10.1109/ICTC.2017.8191011.

[17] N. Dey (Ed.), Advancements in Applied Metaheuristic Computing, IGI Global, 2017.

[18] N. Dey, A. Ashour, S. Bhattacharyya, Applied Nature-Inspired Computing: Algorithms and Case Studies, Springer, 2020.

[19] M. Ebrahimi, E. Shafieibavani, R.K. Wong, C.-H. Chi, A new metaheuristic approach for efficient search in the internet of things, in: 2015 IEEE International Conference on Services Computing, IEEE, 2015, pp. 264–270, http://dx.doi.org/10.1109/SCC.2015.44.

[20] K. Jagatheesan, B. Anand, N. Dey, A.S. Ashour, Effect of SMES unit in AGC of an interconnected multi-area thermal power system with ACO-tuned PID controller, in: Advancements in Applied Metaheuristic Computing, IGI Global, 2018, pp. 164–184.

[21] V. Suryani, S. Sulistyo, W. Widyawan, Trust-based privacy for internet of things, Int. J. Electr. Comput. Eng. (IJECE) 6 (5) (2016) 2396, http://dx.doi.org/10.11591/ijece.v6i5.9678.

[22] J. Kaliannan, A. Baskaran, N. Dey, A.S. Ashour, Ant colony optimization algorithm based PID controller for LFC of single area power system with non-linearity and boiler dynamics, World J. Modelling Simul. 12 (1) (2016) 3–14.

[23] G.N. Nguyen, K. Jagatheesan, A.S. Ashour, B. Anand, N. Dey, Ant colony optimization based load frequency control of multi-area interconnected thermal power system with governor dead-band nonlinearity, in: Smart Trends in Systems, Security and Sustainability, Springer, Singapore, 2018, pp. 157–167.

[24] I. Sabbani, M. Youssfi, O. Bouattane, A multi-agent based on ant colony model for urban traffic management, in: International Conference on Multimedia Computing and Systems, ICMCS, IEEE, 2016, pp. 793–798, http://dx.doi.org/10.11591/ijece.v6i5.9678.

[25] M. Bkassiny, Y. Li, S.K. Jayaweera, A survey on machine-learning techniques in cognitive radios, IEEE Commun. Surv. Tutor. 15 (2013) 1136–1159.

[26] M.H. Adeli, F. Mohammadian, A. Taherpouur, T. Khattab, Optimized collaborative spectrum sensing in energy harvesting cognitive radio networks, in: Proceedings of IEEE wireless conference and networking conference, 2016, pp. 1–7.

[27] S. Althunibat, M. Di Renzo, F. Granelli, Towards energy-efficient cooperative spectrum sensing for cognitive radio networks: an overview, Telecommun. Syst. 59 (1) (2015) 77–91.

[28] M. Song, C. Xin, Y. Zhao, X. Cheng, Dynamic spectrum access: from cognitive radio to network radio, IEEE Wirel. Commun. 19 (01) (2012) 23–29.

[29] G. Wang, C. Guo, S. Feng, C. Feng, S. Wang, A two stage cooperative spectrum sensing method for energy efficiency improvement in cognitive radio, in: IEEE 24th International Symposium on Personal Indoor and Mobile Radio Communications, 2013, pp. 1–5.

[30] J. Wei, X. Zhang, Energy-efficient distributed spectrum sensing for wireless cognitive radio networks, in: INFOCOM IEEE Conference on Computer Communications Workshops, 2010, pp. 1–6.

[31] Kozal A.S., M. Merabti, F. Bouhafs, Spectrum sensing energy tradeoff in multi-hop cluster based cooperative cognitive radio networks, in: 2014 IEEE Conference on Computer Communications Workshops, INFOCOM WKSHPS, 2014, pp. 765–770.

[32] Ouarda Zedadra, Antonio Guerrieri, Antonio Guerrieri, Nicolas Jouandeau, Swarm intelligence-based algorithms within iot-based systems: A review, J. Parallel Distrib. Comput. (2018).

[33] Abdolreza Mohajerani, Davood Gharavian, An ant colony optimization based routing algorithm for extending network lifetime in wireless sensor networks, Wirel. Netw. 22 (2015) 2637–2647.

[34] P. López-Matencio, J. Vales-Alonso, E. Costa-Montenegro, ANT: Agent stigmergy-based IoT-network for enhanced tourist mobility, Mob. Inf. Syst. 2017 (2017) 1–15, http://dx.doi.org/10.1155/2017/1328127.

[35] O. Cosido, C. Loucera, A. Iglesias, Automatic calculation of bicycle routes by combining meta-heuristics and GIS techniques within the framework of smart cities, in: 2013 International Conference on New Concepts in Smart Cities: Fostering Public and Private Alliances, SmartMILE, IEEE, 2013, pp. 1–6, http://dx.doi.org/10.1109/SmartMILE.2013.6708196.

[36] S. Vimal, M. Khari, N. Dey, R.G. Crespo, Y.H. Robinson, Enhanced resource allocation in mobile edge computing using reinforcement learning based MOACO algorithm for IIOT, Comput. Commun. 151 (2020) 355–364.

[37] M.A. Kowshalya, M. Valarmathi, Detection of sybil's across communities over social internet of things, J. Appl. Eng. Sci. 14 (1) (2016) 75–83, http://dx.doi.org/10.5937/jaes14-10176.

[38] J. Meza, H. Espitia, C. Montenegro, R.G. Crespo, Statistical analysis of a multi-objective optimization algorithm based on a model of particles with vorticity behavior, Soft Comput. 20 (9) (2016) 3521–3536.

[39] O. Said, Analysis, design and simulation of internet of things routing algorithm based on ant colony optimization, Int. J. Commun. Syst. 30 (8) (2016) http://dx.doi.org/10.1002/dac.3174.

[40] Y. Jiang, Q. Ding, X. Wang, A recovery model for production scheduling: Combination of disruption management and internet of things, Sci. Program. (2016) http://dx.doi.org/10.1155/2016/8264879.

[41] A. Goli, H.K. Zare, R.T. Moghaddam, A. Sadeghieh, An improved artificial intelligence based on gray wolf optimization and cultural algorithm to predict demand for dairy products: A case study, Int. J. Interact. Multimedia Artif. Intell. 5 (6) (2019) 15–22.

[42] Y. Hong, L. Chen, L. Mo, Optimization of cluster resource indexing of internet of things based on improved ant colony algorithm, Cluster Comput. (2018) 1–9, http://dx.doi.org/10.1007/s10586-017-1496-x.

[43] Sankalap Arora, Satvir Singh, An effective hybrid butterfly optimization algorithm with artificial bee colony for numerical optimization, Int. J. Interact. Multimedia Artif. Intell. 4 (2017) 4–14, http://dx.doi.org/10.9781/ijimai.2017174.

[44] M. Khari, P. Kumar, D. Burgos, R.G. Crespo, Optimized test suites for automated testing using different optimization techniques, Soft Comput. 22 (24) (2018) 8341–8352.

[45] Z. Oralhan, B. Oralhan, Y. Yĭgit, Smart city application: Internet of things (IoT) technologies based smart waste collection using data mining approach and ant colony optimization, Int. Arab J. Inf. Technol. (IAJIT) 14 (4) (2017) 423–427.

[46] C. Bhatt, N. Dey, A.S. Ashour (Eds.), Internet of Things and Big Data Technologies for Next Generation Healthcare, 2017.

[47] A. Schaller, K. Mueller, Motorola's experiences in designing the internet of things, Int. J. Ambient Comput. Intell. 1 (1) (2009) 75–85.
[48] N. Dey, A.E. Hassanien, C. Bhatt, A. Ashour, S.C. Satapathy (Eds.), Internet of Things and Big Data Analytics Toward Next-Generation Intelligence, Springer, Berlin, 2018, pp. 3–549.
[49] G. Elhayatmy, N. Dey, A.S. Ashour, Internet of things based wireless body area network in healthcare, in: Internet of Things and Big Data Analytics Toward Next-Generation Intelligence, Springer, Cham, 2018, pp. 3–20.
[50] L.R. Wenceslao, R.G. Crespo, A.G.C. Sanz, A prototype for linear features generalization, Int. J. Interact. Multimedia Artif. Intell. 1 (3) (2010) 60–66.
[51] G. Mahalaxmi, K.E. Rajakumari, Multi-agent technology to improve the internet of things routing algorithm using ant colony optimization, Indian J. Sci. Technol. 10 (31) (2017) 1–8, http://dx.doi.org/10.17485/ijst/2017/v10i31/112338.
[52] S. Kamel, H. Youssef, Voltage stability enhancement based on optimal allocation of shunt compensation devices using lightning attachment procedure optimization, Int. J. Interact. Multimedia Artif. Intell. 5 (7) (2019) 125–134.
[53] Yuan HongLi, Chen Lianguang Mo, Optimization of cluster resource indexing of internet of things based on improved ant colony algorithm, Cluster Comput. 22 (4) (2019).
[54] H. ElSawy, E. Hossain, M. Haenggi, Stochastic geometry for modeling, analysis, and design of multi-tier and cognitive cellular wireless networks: A survey, IEEE Commun. Surv. Tutor. 15 (2013) 996–1019.
[55] S. Dong, P. Agrawal, K. Sivalingam, Reinforcement learning based geographic routing protocol for UWB wireless sensor network, in: Proc. IEEE Global Telecommunications Conference, GLOBECOM'07, Washington, USA, Jun. 2007, pp. 652–656.
[56] M. Khari, A.K. Garg, R.G. Crespo, E. Verdú, Gesture recognition of RGB and RGB-D static images using convolutional neural networks, Int. J. Interact. Multimedia Artif. Intell. 5 (7) (2019).
[57] A.M. Hemeida, R. Mansour, M.E. Hussein, Multilevel thresholding for image segmentation using an improved electromagnetism optimization algorithm, Int. J. Interact. Multimedia Artif. Intell. 5 (4) (2019).
[58] S.M. Jameii, K. Faez, M. Dehghan, Multiobjective optimization for topology and coverage control in wireless sensor networks, Int. J. Distrib. Sens. Netw. 11 (2015) http://dx.doi.org/10.1155/2015/363815, Article ID 363815.
[59] J.I.R. Molano, J.M.C. Lovelle, C.E. Montenegro, J.J.R. Granados, R.G. Crespo, Metamodel for integration of internet of things, social networks, the cloud and industry 4.0, J. Ambient Intell. Humanized Comput. 9 (3) (2018) 709–723.
[60] S. Vimal, L. Kalaivani, M. Kaliappan, A. Suresh, Xiao-Zhi Gao, R. Varatharajan, Development of secured data transmission using machine learning based discrete time partial observed markov model and energy optimization in cognitive radio networks, Neural Comput. Appl. (2018) http://dx.doi.org/10.1007/s00521-018-3788-3.
[61] Ian F. Akyildiz, Won-Yeol Lee, Mehmet C. Vuran, Shantidev Mohanty, Next generation/dynamic spectrum access/cognitive radio wireless networks: a survey, Comput. Netw.: Int. J. Comput. Telecommun. Netw. 50 (13) (2006) 2127–2159, http://dx.doi.org/10.1016/j.comnet.2006.05.001.